

TRANSFER FUNCTIONS: METHODS FOR QUANTITATIVE PALEOCEANOGRAPHY BASED ON MICROFOSSILS

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1. INTRODUCTION

One of the main avenues of research in paleoceanography is the use of microfossil assemblages for decoding or reconstructing climatic or oceanographic parameters. This requires appropriate statistical methods after the identification of biological remains and count proportions of each taxon, or presence/absence of specific taxa. The various statistical methods that have been developed for the

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difference between $\mathbf{C}$ and its estimate. A backward approach is often more useful because it allows direct calibration of the climate on the assemblages, as follows: $\hat{\mathbf{C}} = \hat{\mathbf{T}}(\mathbf{X})$. Here, $\hat{\mathbf{T}}$ is the transfer function, which can be empirically defined from calibrations using various statistical techniques.

Transfer function models and methods based on calibrations are based on a few assumptions:

- (1) Climate is the ultimate cause of changes in the paleobiological data.
- (2) The ecological properties of the species considered has not changed between the period analyzed and the present time, and the relationship between the species and the climate is thus uniform through time.
- (3) The modern observations contain all the necessary information to interpret the fossil data.

The third assumption originates from the uniformitarian principle. It is implicit in most paleoecological studies, but constitutes an unprovable premise, especially when going back in time in the geological archives. One major consequence of the three assumptions mentioned above is the implicit condition of equilibrium between the present distribution of species and climate.

Since the early work of Imbrie and Kipp (1971), different types of approaches based on multi-variate statistical methods have been developed and widely applied in paleoceanography to reconstruct paleotemperatures. They used primarily planktonic foraminifera, diatoms, and radiolarian assemblages (e.g., CLIMAP, 1976, 1981; Pichon et al., 1992; Koç, Jansen, & Haffidason, 1993; Sabin & Pisias, 1996).

An alternative to the transfer function model and calibration approaches is the analogue-based approach, introduced for the first time in marine sciences by Hutson (1980). It fundamentally differs from transfer function models, as there is no strict calibration. It supposes that a given assemblage of taxa in the fossil record is most likely to have occurred under a combination of environmental conditions characterizing similar modern assemblages of taxa. This approach therefore relies on the comparison between assemblages rather than the direct relationship between taxa and climate or hydrographic parameters. When comparing the assemblages, it is necessary to define an adequate way to evaluate the distance, which quantitatively defines the dissimilarity between the spectra. The analogue-based approach requires the availability of extensive sets of modern assemblages for comparison, especially since there is no calibration allowing extrapolation. Adequate knowledge of the ecology and physiology of the taxa is necessary, not for calibration, but to avoid over-interpreting the data. Two advantages must be pointed out. First, several environmental variables can be reconstructed concurrently, thus taking into account their interactions. Second, no assumption is made about the form of the underlying relationship between taxon abundance or assemblage components and the environmental gradients, unlike calibration approaches that assume linearity or modal distributions. After an early use in paleoceanography by Hutson (1980) and Prell (1985), we had to wait until the 1990s (e.g., de Vernal, Rochon, Hillaire-Marcel, Turon, & Guiot, 1993; de Vernal, Turon, & Guiot, 1994; Waelbroeck et al., 1998) for a revival of the analogue-based approach in paleoceanography.

reconstruction of climate and ocean parameters based on microfossil assemblages are generally referred to as transfer functions in a very large sense. However, the term “transfer function” has a specific meaning from a mathematical point of view, and has different meanings depending on the context. In a paleoclimatic context, the term “transfer function” was apparently used for the first time by Webb and Bryson (1972). In paleoceanography, “transfer functions” may refer to the method developed by Imbrie and Kipp in the early 1970s for the reconstruction of sea-surface temperatures (SSTs) based on multi-variate analyses of planktonic foraminifer assemblages. According to Sachs, Webb, and Clark (1977), the transfer function procedure is defined as a method producing calibrated quantitative estimates of some parameters of the environment, such as seasonal or monthly air or ocean surface temperature, from proxy data such as biological assemblages. By extension, transfer functions may thus include all methods that permit quantitative estimates of ocean or climate parameters using paleontological data.

For a long time, Quaternary paleoecologists have used intuitive methods to reconstruct paleoclimates or paleoenvironments from biological data. The most common approach has been to compare the present-day distribution of selected species with the corresponding distribution of climate variables thought to be determinant for the distribution of the selected species. This method has been called “indicator-species”. It is univariate because the species are analyzed separately and related to one climatic variable. Univariate methods can be the source of a number of problems. In particular, the species respond to a combination of climatic variables and their distribution is controlled by different climatic or hydrographic factors in different parts of their ranges. Moreover, climate parameters are often interrelated. Thus, it has been necessary to develop methods taking into account the ecological complexity of species and assemblages, and of their relationships with climatic factors.

A first attempt was made using a multi-variate indicator-species method, which takes into account several species (Iversen, 1944; Atkinson, Briffa, & Coope, 1987). However, this method still disregards much relevant information concerning species abundances and requires an accurate knowledge and understanding of the present distribution of the species. Such knowledge is possible on land from the observation of the modern vegetation and fauna, but it is much more difficult to develop in marine environments where *in situ* measurements and observations of the phytoplankton and zooplankton are limited in time and space.

A key solution was found with the development of response models and their inverse, the “transfer functions” (Imbrie & Kipp, 1971). These methods assume that each species lives in a given range of climatic and environmental conditions. The problem is having an adequate understanding of the response of plants, animals, or protists to climate and other abiotic and biotic factors. In the case of the response model, the assemblage $\mathbf{X}$ is expressed as a function $\mathbf{R}$ of climate $\mathbf{C}$ and non-climatic factors $\mathbf{D}$ as follows: $\mathbf{X} = \mathbf{R}(\mathbf{C}, \mathbf{D})$. If climate is the main factor controlling the biological response, it is possible to approximate the response function by $\mathbf{X} = \mathbf{R}_c(\mathbf{C})$. The inverse of this response function $\mathbf{X}$ enables one to calculate climatic parameters from known biological responses: $\hat{\mathbf{C}} = \hat{\mathbf{R}}_c^{-1}(\mathbf{X})$. With the exception of a situation characterized by one taxon responding to one climatic factor, the solution for $\hat{\mathbf{R}}_c^{-1}$ can be approached only in a statistical way by minimizing the

In this chapter, we first describe the principles of the different calibration- or similarity-based methods that have been developed for quantitative reconstruction of climate and ocean parameters using biological remains (Table 1). Subsequently, we try to compare the respective use and performance of these methods with an example using a North Atlantic foraminifer data set. In the discussion, we address

Table 1 Summary of the Various Transfer Function Methods used for Paleoclimate or Paleooceanographic Reconstructions Based on Microfossil Data (See Text for More Details).

Type of approach	Family	Sub-family	Method	Order
Calibration	Ordination techniques	Indirect gradient analyses	Regression on principal components (e.g., the Q-mode Imbrie and Kipp method — I&K)	Linear or quadratic
			Regression on correspondence factors (Roux, 1979)	Unimodal
		Direct gradient analyses	Partial least Square (PLS; ter Braak, Juggins, Birks, & Van der Voet, 1993)	Linear
			Canonical correlation (e.g., Guiot, Berger, Munaut, & Till, 1983)	Linear
			Canonical correspondence analyses (e.g., Gasse & Tekaiia, 1983)	Unimodal
			Weighted averaging — Partial least-squares (WA-PLS; e.g., Korkhola & Weckström, 2000)	Unimodal
	Generalized models (GLM)		Logistic regression (e.g., ter Braak & Looman, 1986)	Unimodal
	Non-parametric methods		Generalized additive model (GAM; e.g., Gersonde, Crosta, Abelmann, & Armand, 2005)	Non-linear
			Artificial neural network (ANN; e.g., Malmgren & Nordlund, 1996)	Non-linear
Similarity	Presence/absence		Indicator species method (ISM; Iversen, 1944)	Non-linear
			Mutual climatic range method (MCR; Atkinson et al., 1987)	Non-linear
	Abundances	Unconstrained	Modern analogue technique (MAT; Hutson, 1980)	Non-linear
		Constrained	MAT constrained with external variable (e.g., Guiot et al., 1993)	Non-linear
			MAT constrained geographically (SIMMAX; Pflaumann, Duprat, Pujol, & Labeyrie 1996)	Non-linear
		Response surface	Response surface method (Bartlein, Prentice, & Webb, 1986)	Non-linear
			Revised analogue method (RAM; Waelbroeck et al., 1998)	Non-linear

key issues with respect to limitations and potential future developments. Finally, the applications of different methods using microfossil assemblages for the reconstruction of ocean parameters are briefly presented.

2. METHODS BASED ON CALIBRATION

The transfer function is a predictive method. Whereas response functions aim at describing the relationships between species and abiotic factors, transfer function models aim at making predictions that are as close as possible to reality. The error of prediction is, however, difficult to evaluate and is never known with certainty. It is generally larger than the calibration error, which is obtained by comparing the data used for calibration (also called “training data set”) and the estimates. **The values for the error of prediction can be evaluated by a validation procedure in which part of the modern data available is not used for calibration, and is reserved for prediction tests.** This independent verification is nevertheless very limited (but better than nothing), as nobody knows for certain what is really happening when we move from the domain of modern data towards the domain of fossil data.

2.1. Methods Based on Ordination

Ordination is a general term for multi-variate techniques sorting multi-dimensional data points along one or several axes in order to visualize their structure. In paleo-climatology and paleoceanography, we hope that they are ordered along the environmental or climatic gradients we want to reconstruct. There are different ordination techniques (Table 1), all of which differ slightly with respect to the mathematical approach used to calculate species and sample similarity/dissimilarity. The first family of methods is the indirect gradient analysis, with ordination along hidden variables. It includes the principal component analysis and the correspondence analysis (CA). The second family of methods is the direct gradient analysis. It has the advantage of expressing explicitly the gradient in relation to environmental variables. It includes the canonical analyses and the methods derived from weighted averaging (WA) and partial least square.

Following ter Braak, Juggins, Birks, and Van der Voet (1993), the calibration is done between an environmental vector $\mathbf{y}$ ($y_i, i = 1 \dots n$) and a species matrix $\mathbf{X}$ ($x_{ij}, i = 1 \dots n, j = 1 \dots m$). The subscript i (from 1 to n) is the number of the sites where the m species are recorded (the columns of $\mathbf{X}$) and where the environmental variable are to be reconstructed.

2.1.1. Principal component analysis

The principal components analysis (PCA) is one of the earliest ordination techniques and is the basis of the Imbrie and Kipp method (I&K). It uses a rigid rotation to derive orthogonal axes, which maximize the variance in the species data set. However, it does not guarantee that the variance of the climatic variables is

maximized. It is then accompanied by a multiple regression to project the climate on the species space.

The PCA searches for a weight vector $\mathbf{b}$, which is the first component that maximizes the variance of a linear combination $\mathbf{z}$ of the species, subject to the condition that the norm of $\mathbf{b}$ is 1. The quantity V to be maximized is:

$$V = \mathbf{z}'\mathbf{z}/\mathbf{b}'\mathbf{b} \text{ where } \mathbf{z} = \mathbf{X}\mathbf{b} \text{ (}\mathbf{z} \text{ being centered and normalized)} \quad (1)$$

In such an analysis, the species are first standardized, i.e., the columns of $\mathbf{X}$ have a zero mean and unit variance. The PCA can be done in R-mode, with $\mathbf{b}$ being an eigenvector of $\mathbf{X}'\mathbf{X}$ (which is proportional to the correlation matrix of the species). Alternatively, the PCA can be made in a Q-mode, and $\mathbf{b}$ is then an eigenvector of $\mathbf{X}\mathbf{X}'$ (which measures the association between the sites). In both modes, the maximum number of principal components is equal to the minimum of m (number of species) and n (number of sites). Equation (1) defines the first principal component. The second, third and following components are defined in the same way, with the condition that the components are all independent. The estimate of $\mathbf{y}$ is finally obtained by regression on these components $\mathbf{z}$.

It is important to mention here that several terminologies are used for principal component analyses depending upon the application domain. Imbrie and Kipp (1971) used the term factor analysis in reference to the human sciences, which introduced the expression after identification of latent factors in data sets. In these analyses, a small number of factors yielding a clear interpretation are kept, whereas the remaining ones are considered to represent noise. In climatology, the term empirical orthogonal function (EOF) is often used because the focus is to detect independent patterns typical in climatic fields of research. All of these techniques use eigenvectors analyses of a covariance or correlation matrix and are mathematically identical. The term “factor analysis” has only been used in the recent literature to refer to the CA (Section 2.1.3). Here, we will continue to use PCA, although Imbrie and Kipp used the expression “factor analysis”.

Imbrie and Kipp (1971) have developed a procedure in three steps. The first step consists of estimating the physical variables to describe the biotic assemblages in core-tops using a PCA, which is an objective way of clustering different species into assemblages of species. The PCA used is a Q-mode on non-centered percentages. To facilitate the interpretation, they used a varimax rotation, although it was not necessary to improve the performance of the method. A small number of components usually represent a large proportion of the variance. The components retained are related to typical assemblages, such as arctic, subarctic, subtropical and tropical. The second step leads to the derivation of a relationship between each environmental variable and these components using multiple (linear or quadratic) regressions (possibly stepwise regression). The third step projects the down-core data on the principal components and thus allows for the calculation of factor scores for each sample. In the final step, the scores are entered into the calibration equations to calculate paleotemperatures.

CLIMAP (1976, 1981) have adopted this method in their worldwide reconstruction of the last glacial maximum (LGM) SST. This reconstruction has been often used by climate modelers as a boundary condition for paleoclimate

simulations of the LGM (e.g., Joussaume & Taylor, 1995; IPCC, 2001; Braconnot, 2004). Koç and Schrader (1990), Koç et al. (1993), Andersen, Koç, Jennings, and Andrews (2004) and De Sève (2005) have applied the method to marine diatoms in northern Atlantic regions. They have been able to interpret six factors explaining more than 90% of the species variance. Gersonde et al. (2005) have also used this technique, among two others, to estimate the climate at the LGM in the Circum-Antarctic Ocean based on siliceous microfossils. Sabin and Pisias (1996) and Pisias, Roelofs, and Weber (1997) have developed and applied the Imbrie and Kipp (1971) technique for the reconstruction of SSTs in the North Pacific based on radiolarian assemblages. Mix, Morey, and Pisias (1999) have modified the way with which to calculate the PCA in order to solve the problem of no-analogues in the LGM: instead of calculations based on modern data, they made calculations of the PCA on a large set of LGM data. Such a variant in the approach seems to be a suitable way to make progress in spite of the limitations resulting from a no-analogue situation.

2.1.2. Partial least square analysis and canonical correlation analysis

While the PCA does not try to fit at best the predictand $\mathbf{y}$ (recall that a predictand is a variable to predict, here the climate), the partial least square analysis (PLS) maximizes the covariance between the predictand and a linear combination of the species subjected to the condition that the norm of $\mathbf{b}$ is equal to 1. The quantity C to be maximized for the first component is

$$C = \mathbf{z}'\mathbf{y}/\mathbf{b}'\mathbf{b} \text{ where } \mathbf{z} = \mathbf{X}\mathbf{b} \quad (2)$$

$\mathbf{z}$ being a linear combination of the species abundance, the analysis aims at finding a linear combination to maximize the covariance with the environmental parameter. As in PCA, the following components must be orthogonal to the first one and together. In this method, the interpretation of the components is linked to the predictand to be reconstructed; whereas in the PCA, the components must be calibrated *a posteriori* on the predictor (a predictor is a variable used to predict an dependent variable, they are here the taxa percentages). There are two possibilities of generalization to several predictands:

- (1) One is to maximize the sum of the correlations of each predictand. This is the principal component analysis of instrumental variables (Sabatier & Van Campo, 1984) or redundancy analysis (ter Braak & Prentice, 1988).
- (2) The second possibility is to maximize the correlation between a linear combination of the predictors and a linear combination of the predictands. This allows for the deduction of the relationships of the predictands in relation to the predictors (Fritts, Blasing, Hayden, & Kutzbach, 1971; Guiot et al., 1983).

2.1.3. Correspondence analysis

With species assemblages, CA (Benzécri, 1973; Hill, 1974) is often more suitable than PCA because unimodal relationships replace linear relationships between species and climate parameters (ter Braak, 1985). However, the CA requires the introduction of norms. Let us define $\mathbf{R}$ as the diagonal matrix $(x_1, x_2, \dots, x_n)$

where x_i is the total abundance of all species at site i (it is equal to 100 in the case of percentages). $\mathbf{K}$ is the diagonal matrix $(x_{.1}, x_{.2}, \dots, x_{.m})$ where $x_{.j}$ is the total abundance of species j at all sites. CA searches for a centered weight vector $\mathbf{u}$ that maximizes $\mathbf{V}$:

$$\mathbf{V} = \mathbf{z}'\mathbf{R}\mathbf{z}/\mathbf{u}'\mathbf{K}\mathbf{u} \text{ where } \mathbf{z} = \mathbf{R}^{-1}\mathbf{X}\mathbf{u} \quad (3)$$

The following components are defined in the same way with the additional condition that they are orthogonal. The predictand $\mathbf{y}$ is estimated by regression on the components $\mathbf{z}$. Roux (1979) proved the superiority of this method by its application on the data set of Imbrie and Kipp (1971). As for PCA, this method imposes the projection of $\mathbf{y}$ on the components by regression. This two-step method is also known as an indirect gradient analysis (ter Braak, 1986). Detrended correspondence analysis (DCA) corresponds to CA corrected for the arch effect (when a distribution of sites along an arch is observed after plotting of their scores on axes 1 and 2).

2.1.4. Weighted averaging partial least square regression (WA-PLS)

The WA-PLS is related to the CA method as the PLS is related to the PCA (ter Braak et al., 1993). WA-PLS searches for a vector $\mathbf{u}$ with weighted norm equal to 1 that maximizes C :

$$C = \mathbf{z}'\mathbf{R}\mathbf{y}/\mathbf{u}'\mathbf{K}\mathbf{u} \text{ where } \mathbf{z} = \mathbf{R}^{-1}\mathbf{X}\mathbf{u} \quad (4)$$

The following components are found in the same way as in the previous analyses, but with the additional condition that they are R-orthogonal. A particular case of this analysis is the WA method where only one component is retained. The score u_k is then given by the average of the environmental parameters on the n observations weighted by the abundance of the k th species (this corresponds to the first factor or component). The score obtained gives an indication of where the optimum of species k occurs in the environmental domain, and $\mathbf{y}$ is reconstructed by averaging the m values of u_k weighted by the species abundances in the analyzed sample (this corresponds to the regression). This method is efficient for reconstructing chemical parameters from lake diatoms (ter Braak & Juggins, 1993). The generalization of several predictands is provided by canonical correspondence analysis (CCA) (ter Braak, 1986, 1987; Gasse & Tekaia, 1983).

WA-PLS is commonly applied in paleolimnology and paleoclimatology by using lake diatoms (Racca, Gregory-Eaves, Pienitz, & Prairie, 2004; Korkhola & Weckström, 2000) and pollen (Seppä, Birks, Odland, Poska, & Veski, 2004). It has been applied in paleoceanography for the reconstruction of SSTs north of Iceland during the late Holocene based on diatoms (Jiang, Seidenkrantz, Knudsen, & Eiriksson, 2002). WA-PLS has also been used for SST reconstruction at LGM in the Pacific and Atlantic Oceans based on planktonic foraminifera by Morey, Mix, and Pisias (2005). In their paper, the authors provide a very careful description of the CCA approach showing its relationship with CA.

2.2. Generalized Models and Non-Parametric Methods

2.2.1. General linear model

The general linear model (GLM) can be seen as an extension of the linear multiple regression:

$$\mathbf{z} = \mathbf{X}\mathbf{b} = \sum_{j=1}^m b_j x_j$$

$$\mathbf{y} = \mathbf{z} + \mathbf{e} \quad (5)$$

where $\mathbf{y}$ is the climatic variable, $\mathbf{z}$ its estimation, $\mathbf{X}$ the species matrix and $\mathbf{b}$ the vector of regression coefficients. A major limitation of the simple linear model is met when the $\mathbf{X}$ variables are not linearly independent. In such cases, the use of principal components of $\mathbf{X}$ solves the problem. An advantage of GLM is the possibility of using categorical variables and giving access to multi-variate tests for assessing the significance of the model. The basic idea of GLM is that the covariance matrix of the residuals is not required to be a diagonal matrix with all the terms equal to σ_e^2 with a Gaussian distribution. We may impose any sort of distribution, in particular a binomial distribution, and then define a logistic model (ter Braak & Looman, 1986). We may also weight the observations in the fitting process in order to emphasize some classes of observations.

The species assemblages have m variables, but the sum of the percentages is 100. Therefore, when $m-1$ species are known, the m th is fixed and one regressor is redundant, and we can remove it. However, because the aim is not only to reconstruct the climatic variable $\mathbf{y}$ but also to analyze how the species respond to climate, we do not want to remove any regressor arbitrarily. GLM allows the possibility of retaining all of the regressors in the analysis.

When categorical variables are added to the model, there are two ways of coding them. The first way is to code each category of the predictor by one numerical value (1, 2, ...), thus adding only one predictor. The second way is to define a new binary predictor (taking the value of 1 or 0 depending on whether the category is realized or not) for each category, thus adding a large number of predictors for a single additional variable. GLM can be used with both coding methods. In transfer function models, a categorical variable could, for example, be the membership to a given biome (Guiot et al., 1993).

2.2.2. Generalized additive model

The generalized additive model (GAM) (Hastie & Tibshirani, 1990) extends traditional regression as effectively as GLM by replacing $\sum_{j=1}^m b_j x_j$ in Equation (5) by $\sum_{j=1}^m b_j f_j(x_j)$. Since the functions f_j can be non-linear, GAM belongs to a non-linear family of methods. It is also a non-parametric method in the sense that it is not necessary to know the analytical form of f_j . Usually, the function used is a spline function, which is very flexible and makes this model very efficient.

The non-linear modeling procedures are useful for two reasons. First, they help to prevent model misspecification, which can lead to incorrect conclusions regarding treatment efficacy. Second, they provide information about the relationships

between predictors and predictands that is not shown by standard modeling techniques. Nevertheless, the counterpart is a risk of over-parameterization and the difficulty of understanding non-analytical functions.

The locally weighted regression proposed by Cleveland and Devlin (1988) fits a surface to the data in a multi-dimensional space by using the family of the spline functions. It has the advantage of being non-parametric and does not require formal specification of the surface function. Gersonde et al. (2005) used GAM to estimate the sea-ice concentration from diatom assemblages during the LGM. This family of methods is used to estimate the response of species to several environmental variables, as seen in the response surface section.

2.2.3. Artificial neural networks

The artificial neural network (ANN) seems to be an excellent approach for a large class of non-linear problems (Walczak & Wegscheider, 1994). ANN implies a black-box approach, which allows very close data fitting, although it is hard to analyze the causality between input and output variables. An ANN consists of a number of simple and interconnected processors called neurons. A signal coming from input variables, the species assemblages, passes through these neurons to reach the output variables, the environmental variables. The interconnections are defined by coefficients that are iteratively tuned to best fit the output variables. In the most popular case, the architecture of the ANN is composed of one hidden layer with a number of neurons, depending on the complexity of the problem. At each neuron, the incoming signal is transformed by a non-linear function (called activation function) and sent to the next layer. An example of possible activation function is the sigmoid function, which translates the input in the $[0,1]$ domain and attenuates extreme values. This function introduces non-linearity in the model. The process of estimating the adjustable coefficients is called “training”. Several iterations are required for this process. One of the most commonly used ANN methods is the feed-forward network trained with the back-propagation learning algorithm.

One major issue with ANN is the difficulty to find the proper structure of the network (number of neurons and hidden layers). There is no other method than to try different designs and to select the most satisfying one. Another issue is the number of iterations needed during the training for correct calibration and reconstruction. It is indeed important to stop the iterations before overestimation. Because of its non-linear property, ANN may fit the training data set very well, including the noise inherent to any data set, which can restrict the ability of making correct predictions outside of the range of the training. To avoid such a problem, it is necessary to divide the training data set into two subsets where one is used for calibration and the other is used for independent verification: when the verification error increases and becomes too large, the iterations have to be stopped. This is called early stopping.

This method has been introduced in continental paleoclimatology by Guiot, Cheddadi, Prentice, and Jolly (1996) and in paleoceanography by Malmgren and Nordlund (1996). Peyron et al. (1998) have shown its potential in reconstructing LGM climate from pollen data by means of reducing the number of predictors by grouping them into plant functional types. Such grouping is motivated by the fact

that a large number of predictors leads to a much larger set of parameters and finally induces a risk of over-parameterization. Guiot and Tessier (1997) and Guiot et al. (2005) have shown the advantage of ANN for extracting climatic signals from tree-ring data. Racca, Philibert, Racca, and Prairie (2001) have shown the better predictive power of ANN over WA and WA-PLS with lake diatoms, but stated that a multi-model approach is better for capturing information more accurately from diatom assemblages.

3. METHODS BASED ON SIMILARITY

Instead of calibrating a relationship between climate and species, this family of methods (Table 1) is based on the comparison of past assemblages to modern assemblages. The mutual climatic range (MCR) methods (and their predecessor, the species indicator method) use maps of the modern distribution of species and consider only the occurrence in terms of presence/absence. The analogue methods directly or indirectly compare fossil assemblages to modern assemblages using a similarity index.

3.1. Mutual Range Method

The indicator species method (ISM) is based on a set of distribution maps showing the modern geographical range of taxa and records of the environmental or climatic variables that determine their range. Then, the distribution of the taxa is re-mapped into a space defined by axes, which are the environmental or climatic variables to be reconstructed. Iversen (1944) and Hintikka (1963) presented early examples of such re-mapping when they used records of the presence or absence of higher plant taxa in the proximity of meteorological stations to draw scatter plots showing the occurrence of taxa with respect to summer and winter temperatures.

Atkinson et al. (1987) generalized ISM to the MCR method dealing with a large number of taxa. Current reconstruction methods generally employ three or four environmental variables concurrently to define a three- or four-dimension environmental space. A regular grid is laid out in this space and the data are used to determine the status of the taxon at each grid point. From the re-mapped data, a climatic envelope that encompasses the environmental space of taxon occurrence is mathematically defined. In order to make reconstructions using MCRs, it is necessary to search for grid points at which the combination of taxa found in the fossil assemblage may co-occur. Although this method is straightforward in principle, problems arise from co-occurrence of taxa that do not have MCRs in the modern data set. In such circumstances, the best solution is to find the least dissimilar combination(s) of taxa. The reconstructed values given by the MCR method are ranges for each of the environmental variables. In general, these ranges are smaller for more diverse assemblages. However, in assemblages of low species diversity, the presence of a taxon characterized by an extremely narrow range of occurrence may reduce the range of possible environmental conditions.

This method has been criticized because it is based only on presence/absence of species, while abundances of taxa are often available. In some situations, when there are assemblages that no longer exist in present times, abundances cannot be used as with other methods. For example, in the Pliocene, fossil pollen assemblages from Mediterranean areas resemble modern ones but include Chinese and North American pollen taxa. In this case, the MCR approach is the most suitable. It has been used by Fauquette, Guiot, and Suc (1998) with some additional information such as absence/presence/strong presence. Klotz, Guiot, and Mosbrugger (2003) have improved the method by implementing a probability approach, which is more subtle than the common MCR based exclusively on binary information. With some proxies, such as plant macrofossil or mollusk remains, the abundance in the sediment is not necessarily a direct function of the actual abundance of the organism in the environment. In these cases, the MCR or its variants are more suitable for climate reconstruction than other methods (Moine, Rousseau, Jolly, & Vianey-Liaud, 2002; Pross, Klotz, & Mosbrugger, 2000). MRC has not been applied to marine data, but its recent development may offer sufficient improvement to motivate applications, notably for periods and regions where modern analogues do not exist.

3.2. Modern Analogues Method

The modern analogue technique (MAT) is based on the collection and analysis of a large number of modern or surface samples representing the same type(s) of sedimentary environment as the fossil samples from which microfossil assemblages have been recorded. These samples provide point data and thus contrast with the distribution maps forming the basis of the mutual range method, which gives the illusion of spatially continuous information. With MAT, environmental data must be interpolated at the sampling sites, as is the case for the calibration methods. Fossil assemblages may contain combinations of taxa that respond to different aspects of the environment. In such cases, it might be suitable to weight the taxa according to their sensitivity against the environmental variables to be reconstructed. The simplest weighting is the exclusion from the assemblages of taxa that do not display a response to any of the environmental variables being reconstructed. Another method of weighting is given by paleobioclimate indicator loadings (Guiot, 1990). The taxa weights are derived by means of an ordination of the fossil data: scores are obtained for a series of ordination axes and these axes are examined in order to determine which one is best related to the taxa showing a strong response to the environmental variables to be reconstructed. The scores for this axis are then used as weights when assessing the degree of analogy. Thus, the most responsive taxa have the greatest influence upon the choice of analogues.

An appropriate distance measure is required in order to evaluate the degree of analogy between fossil and modern samples. In most cases, MAT is applied to microfossil data that are expressed as percentage values. It is possible to use a variety of distance measures with percentage data, especially with data transformations reducing the influence of the ubiquitous taxa that are often more abundant in the assemblages. The most widely used distance measure is known as the “chord

distance”, which is particularly effective with percentage data (Overpeck, Webb, & Prentice, 1985) and may be computed using the formula below:

$$d_{ij}^2 = \sum_{k=1}^m \left(\sqrt{p_{ik}} - \sqrt{p_{jk}} \right)^2 \quad (6)$$

where m is the number of taxa, p_{ik} (resp. p_{jk}) is the proportion of taxon k in fossil assemblage i (resp. modern assemblage j) and d_{ij}^2 the squared chord distance between assemblages j and i .

The smaller the chord distance, the greater is the degree of analogy between the two samples. Thus, paleoenvironmental reconstructions are based on a small number of modern samples that are the most similar to the fossil sample. The dispersion of the analogues around the reconstructed value provides a partial estimate of the uncertainty, which is related to the tolerance of the proxy in a relatively large range of climatic or oceanographic conditions. However, this estimate does not include any uncertainty related to the fitting error between the climatic or oceanographic conditions. A solution has been proposed by Nakagawa, Tarasov, Nishidac, Gotandaa, and Yasudaa (2002), who applied MAT to meteorological station data and calculated a linear relationship between observations and reconstructions with confidence intervals. For a fossil assemblage, this relationship can be applied to the reconstruction from the MAT, providing the required error bar. Another issue in estimating the uncertainty depends upon the number of spatially close analogues available for a given sample. A way to remedy this problem is to do an even re-sampling of the modern assemblages similar to that of the response surface approach (*cf.* Section 3.3). The determination of the number of analogues to be taken is also critical. Waelbroeck et al. (1998) proposed selecting the set of analogues before a sharp increase of the distance; Gavin, Oswald, Wahl, and Williams (2003) proposed another statistical tool named the “receiver operating characteristic” curve which helps to decide the threshold between analogue and no-analogue spectra.

MAT was used for the first time in paleoceanography by Hutson (1980). It was applied by Prell (1985) on the CLIMAP (1976, 1981) data. de Vernal et al. (1994, 2001, 2005) have shown its ability to reconstruct past SST, salinity and ice cover in subpolar seas from dinocyst assemblages. It is now considered to be a standard method, which is often compared to other ones for assessing the reliability of reconstructions based on other methods (e.g., Kucera et al., 2005b; Barrows & Juggins, 2005; Chen, Huang, Pflaumann, Waelbroeck, & Kucera, 2005; Gersonde et al., 2005).

3.3. Constrained Analogues

A large range of analogues may reduce the possibility of the selection of “wrong analogues”. However, the problem of wrong analogues remains acute in the case of fossil assemblages characterized by a limited number of taxa, which are ubiquitous and represent a wide range of environmental conditions. When additional qualitative or quantitative information is available, constrained-analogue methods have

shown great promise with pollen data. Two main approaches can be used. One consists of applying a constraint for the selection of analogues in a given biome: all of the assemblages from other biomes are rejected even if the distance with fossil assemblage is very low. The second approach to constrain the choice of analogues uses paleoclimatic information derived from independent proxies. For example, fossil beetle assemblages (Guiot et al., 1993), sedimentological indices (Seret, Guiot, Wansard, Beaulieu, & Reille, 1992) and past lake levels (Cheddadi, Yu, Guiot, Harrison, & Prentice, 1997) have been used together to constrain the choice of analogues in pollen-based paleoclimate reconstructions. The advantage of this procedure is the possibility of reducing the confidence interval of the climatic parameter common to the various proxies and the opportunity to reconstruct additional climatic parameters. For example, constraints on pollen analogues derived from fossil beetle assemblages may be used to improve the precision of July temperature reconstructions, and the resulting constrained pollen analogues also yield more coherent winter temperature estimates. The principal disadvantage of the “multi-proxy constraint” approach is that independent paleoclimate inferences from alternate proxies are no longer available for comparison. Nevertheless, if the climatic control and sensitivity of the various proxies is well understood, the “multi-proxy constraint” approach can yield internally consistent reconstructions of paleoclimate parameters, including seasonality of temperature and precipitation.

In addition to the distance between spectra, Pflaumann et al. (1996) have introduced a constraint related to the geographical distance: the analogues chosen are those located at the smallest geographical distance from the fossil site (SIMMAX method). It is a MAT constrained by geography. Such an approach could be useful for warm periods characterized by fossil assemblages similar to the modern ones, but it is difficult to justify for glacial periods because the best analogues can be located at a very large distance from the study site. For example, Malmgren, Kucera, Nyberg, and Waelbroeck (2001) have shown that the results do not match the ranking obtained with the calibration data set when a geographical constraint is applied to an independent validation data set and to an additional fossil data set. Therefore, this approach is not further considered here.

3.4. Response Surfaces

The first step of the response surface analogue technique (RSAT) is a re-mapping of the taxa into an environmental space using the modern data set (and not biogeographical distribution of modern species as in the MRM approach). The main method of this type currently applied is the quantitative environmental response surfaces (Bartlein et al., 1986; Huntley, 1994). In this method, the surfaces are fitted to the abundance values of each taxon used for reconstructions. The abundance of the taxon at each grid point can be fitted either by a flexible extrapolation technique such as the locally weighted regression (Cleveland & Devlin, 1988) based on tri-cube weighting function (which is related to the GAM approach), or by an interpolation weighted inversely to the distance. The latter is the approach adopted by Waelbroeck et al. (1998) for SSTs estimates based on planktonic foraminifers.

Having fitted response surfaces for the suite of taxa used in reconstructions, the values for the taxa at each grid point can be combined to provide a series of assemblages that may be considered as potential analogues. Thereafter, the procedure is very similar to that of MAT, with 5–10 closest matching assemblages selected for each fossil assemblage and used to provide a reconstruction and a confidence interval. The degree of analogy is once again conventionally measured using the chord distance.

In the RSAT approach, the variation of taxa with distributions independent of climate is smoothed and thus removed by the procedure used to fit the surface. Therefore, if the principal cause of variability of a taxon is non-climatic, the response surface is flat and the taxon contributes little to the choice of analogues. As the new potential assemblages are sampled on a grid, each type of climate is evenly represented, which improves the value of the confidence intervals. These two features represent the greatest advantages of this method.

In paleoceanography, response surfaces have been used to generate new gridded assemblages and combined with original assemblages to fill the parts of the climatic space not covered by data. The method proposed by Waelbroeck et al. (1998), which is called the revised analogue method (RAM), uses gridded assemblages in addition to raw assemblages in order to provide a better shaped data set. The gridded samples are obtained by interpolation in the climatic space, which is more flexible than a surface (Figure 1B). Another modification is made by the introduction of a threshold for the selection of the analogues at the place where a sharp increase of the distance is observed (see Section 3.1). The use of RAM allows for improvements in the reconstruction of SSTs at cold and warm ends of the temperature range. It has been used together with MAT, SIMMAX and ANN in the Multiproxy Approach for the Reconstruction of the Glacial Ocean (MARGO) initiative to reconstruct LGM SST (Barrows & Juggins, 2005; Chen et al., 2005; Hayes, Kucera, Kallel, Sbaiffi, & Rohling, 2005; Kucera et al., 2005b). A more formal approach, using a Bayesian probabilistic approach, is proposed by Haslett et al. (2006).

4. COMPARISON OF METHODS WITH A WORKED EXAMPLE

In this section, most of the methods reviewed above (*cf.* Table 1) are compared from their application to the same data set. For this purpose, we used the MARGO planktonic foraminifer core-top data set from the Atlantic Ocean (e.g., Kucera et al., 2004). It is composed of 862 spectra of 26 taxa (percentages) and the corresponding SSTs in winter (January–February–March) and in summer (July–August–September) is compiled from the 1998 version of the World Ocean Atlas (NODC, 1998). From this data set, we have removed the 26 spectra with winter SSTs $< -1^{\circ}\text{C}$ representing very cold conditions in the North Atlantic, and the 23 spectra with summer SSTs $> 29^{\circ}\text{C}$ representing very warm conditions. These two subsets of data represent extreme conditions at the limits of the main data set and will be used to evaluate how each method performs in almost no-analogue conditions. The main data set contains 813 samples that are randomly divided in one calibration data set of

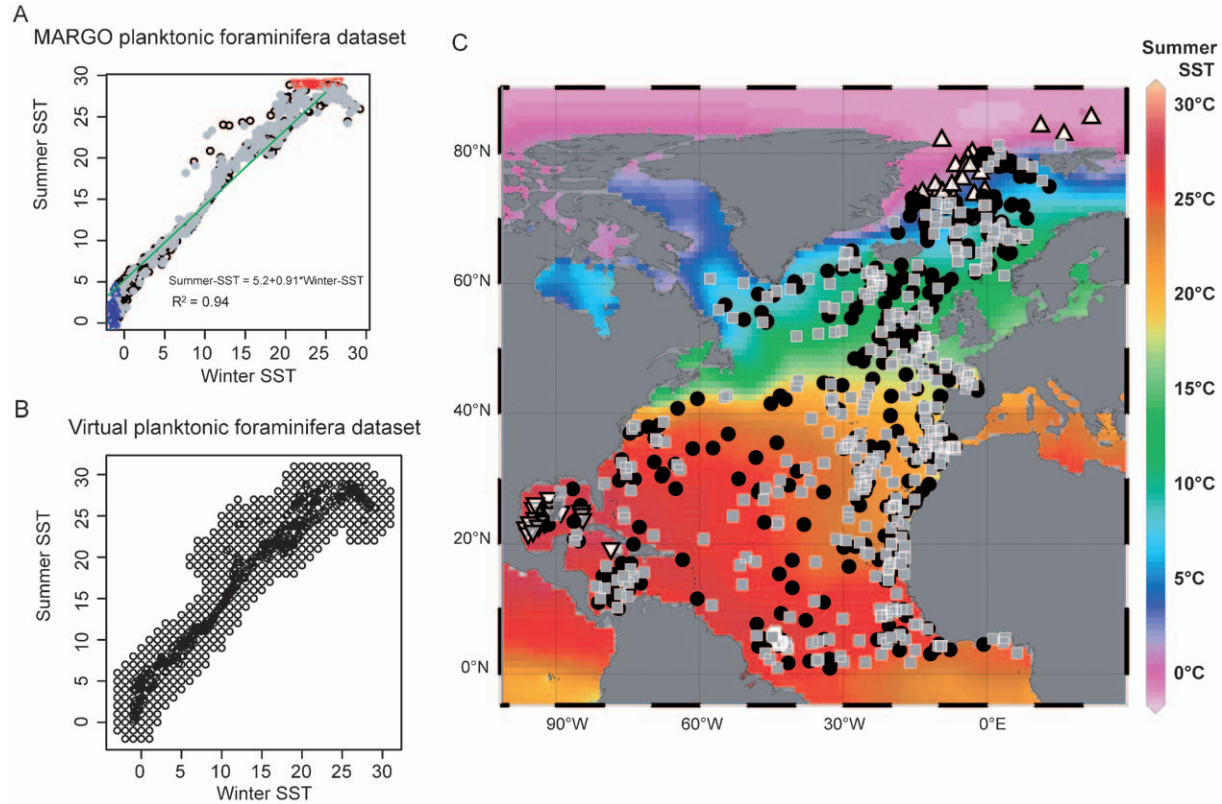


Figure 1 Hydrographical and geographical distribution of samples used in the worked example. (A) Summer vs. winter SSTs at selected sites: the calibration and verification data sets are shown by circles, and the cold and warm sub-sets of data are illustrated by blue and red triangles, respectively. (B) Summer vs. winter SSTs for the virtual planktonic foraminifer data set, established from the interpolation of the climatic space by defining a limit of 3°C for the distance between grid points and raw data. (C) Map of the North Atlantic Ocean showing the summer SST and the location of surface sediment samples: grey squares and black dots are the calibration and verification samples, respectively; the open triangles correspond to the cold samples (Greenland Sea and Arctic Ocean) and the warm samples (Caribbean Sea).

407 samples and a verification data set of 406 samples. Figure 1A, C show the distribution of the samples in the hydrographic and geographic space, respectively. The distribution along the main diagonal is typical of open ocean where there is a high correlation ($r = 0.97$) between winter and summer SSTs.

To assess the prediction error of the various techniques, the root-mean-square error (RMSE) is computed as the square root of the sum of the squared differences between the observed and the predicted values divided by the number of observations. When it is applied to the verification set, the RMSE is called the root-mean-square error of prediction (RMSEP).

All the methods have been implemented in the *R environment for statistical computing and graphics* (<http://www.r-project.org/>). A script called “transfer.r” has been used to implement the eight methods on the worked example with text and graphics outputs. It can be found at the URL: <http://servpal.cerege.fr/paleoceanotransfer/>

Using the I&K method, we reduced the calibration matrix of species relative abundances to eight principal components. Imbrie and Kipp (1971) used the varimax technique to rotate the selected components in order to facilitate the interpretation of the variables. We did not apply this technique because it has no effect on the statistical performance of the reconstruction but only on the interpretation of the biogeographical patterns. The scores of the principal components were used as predictors of the winter and summer SSTs. A curvilinear multiple regression was performed between the two predictands and the eight principal components and their interactions ($PC1 \times PC2$, $PC1 \times PC3 \dots$, $PC2 \times PC3 \dots$). Between 95% and 96% of the SST variance is explained by these predictors (Table 2). In parallel, verification samples were projected on the eight components and their scores were introduced into the calibrated regression. The RMSE was $\sim 1.9^\circ\text{C}$, and the RMSEP on the verification data set was $\sim 2.0^\circ\text{C}$ and 2.2°C for summer and winter, respectively. The RMSEP on the “cold” samples is clearly larger (2.5°C and 4°C); the method cannot yield SST estimates lower than $+1^\circ\text{C}$ in winter and $+4^\circ\text{C}$ in summer (Figure 2). The performance for the “warm” samples was better, especially in winter (RMSEP = 1.5°C), but more dispersed values than found in reality were obtained for summer SSTs (RMSEP = 2.16°C ; Figure 2b).

For the PLS method, we used the kernel algorithm with a leave-one-out cross-validation procedure to select the number of components to be included in the regression. The number of components selected is the one providing the minimum RMSEP on the cross-validation samples (it must be noted that this cross-validation is restricted to the 407 samples calibration data set, the verification data set being exclusively kept for an independent test). For winter SST, we used 13 components explaining 98% of the predictand variance, and for summer SST, we also used 13 components explaining 97% of the variance. The RMSE was lower than for the I&K method (Table 2). The RMSEP calculated on the 406 verification samples were also much lower. With respect to the samples from the cold and warm extremes, the performances were better than those of the I&K method; the winter SST estimates for the cold samples were better, and had values of $\sim 0^\circ\text{C}$ (Figure 2a), and the summer SST estimates for the warm samples were less dispersed, although slightly underestimated in comparison with the I&K estimates (Figure 2b).

Table 2 Performance of the Various Transfer Function Methods of Paleoclimate Reconstructions Applied to Planktonic Foraminifer Data.

Method	R^2 calib.	RMSE	RMSEP-406	RMSEP-cold	RMSEP-warm
Winter SST					
IK	0.96	1.88	2.05	2.48	1.49
PLS	0.98	1.38	1.48	2.05	1.54
WA-PLS	0.98	1.34	1.50	2.05	0.99
GLM	0.97	1.46	1.61	1.07	1.24
GAM	0.98	1.34	1.35	1.70	2.12
ANN	0.98	1.13	1.29	1.74	1.08
MAT	0.99	1.07	1.06	1.43	1.26
RAM	0.99	1.05	1.16	1.24	0.82
Std. dev.		8.95	8.43	0.32	1.14
Summer SST					
IK	0.95	1.95	2.22	3.98	2.16
PLS	0.97	1.49	1.57	3.48	1.80
WA-PLS	0.97	1.50	1.57	3.44	1.07
GLM	0.96	1.68	1.73	2.05	1.64
GAM	0.97	1.40	1.50	3.00	1.16
ANN	0.98	1.19	1.28	3.00	1.84
MAT	0.98	1.17	1.14	2.88	1.01
RAM	0.98	1.22	1.22	2.86	0.81
Std. dev.		8.37	7.51	1.75	0.07

Notes: The coefficient of determination (R^2) and the root square of the mean squared errors (RMSE) are calculated on the calibration data set (407 samples), the root square of the mean squared errors of the prediction (RMSEP) are calculated on the validation data set (406 samples), on the cold data set (26 samples with winter SST < -1°C) and on the warm data set (23 spectra with summer SST > 29°C). The best performing values are enhanced in bold.

For the WA-PLS method, we followed the method proposed by [ter Braak et al. \(1993\)](#) who demonstrated that the solution of WA-PLS can be obtained from a PLS algorithm by pre-processing the taxa and the environment matrices, and by applying the PLS algorithm to the transformed matrices. The leave-one-out cross-validation test led to the retention of 15 and 7 components for winter and summer, respectively. The percentage of variance reconstructed and the RMSE are quite similar to those obtained with the PLS approach, but the RMSEP on the warm samples gives much better results (~1°C; [Table 2](#)), with estimates more closely concentrated along the diagonal than with the PLS method ([Figure 2a, b](#)).

For the GLM method, the predictors are the 26 taxa and the error distribution was assumed to follow a Gaussian law. Because the objective was to focus on the low temperatures, a weighting function equal to the square of the inverse of the SST (corrected to avoid zero values) was defined. For example, SSTs < 0°C are weighted > 1 and SSTs > 20°C are weighted < 0.0025. This arbitrary weighting procedure was applied to illustrate how a GLM performs by focusing on a given environment. The results were similar to those obtained with the WA-PLS, except

that the cold samples yielded a RMSEP half as large, and allowed winter SST estimates of -1°C for observations ranging from the freezing point (-1.8°C) to -1°C (Figure 2). The improvement for summer SSTs was also significant, with estimates of $\sim 2^{\circ}\text{C}$ for observations ranging between -1.8°C and 4.4°C . For the warm samples, the performance was intermediate between those of the PLS and WA-PLS methods.

For the GAM method, the predictors are the first six factors CA extracted from the taxa, and Gaussian law is assumed for the residuals. The six predictors were smoothed by a thin plate regression spline. The multi-dimensional surface was fitted to SST in the six-dimension space. The performance on the calibration samples is similar to that of the previously described methods. The RMSEP of winter SST is larger than that of GLM for both cold and warm samples, but the performance for estimating summer SSTs of warm samples was excellent (Figure 2).

For the ANN method, the size of the species data set was also reduced by using the first ten correspondence factors. We used the architecture of a back-propagation network as described by Malmgren et al. (2001) and Peyron et al. (1998). The number of neurons was set at four. Thus, the size of the input layer was 10, that of the hidden layer size was equal to four and the size of the output layer corresponded to the number of predictands, i.e., two (winter and summer SSTs). During the iterative estimation of the ANN, we regularly calculated the calibration error (on the 407 samples of the calibration data set) and the verification error (on the 406 samples of the verification data set). As expected, the calibration error tended to continuously decrease but the verification error, after an initial decrease, tended to slightly increase. An increase that is too high indicates an over-parameterization. After $\sim 1,000$ iterations, the convergence was considered satisfactory because the verification error remained similar to the calibration error. The RMSE was narrower than with the other methods and the RMSEP belongs to the group of methods that performed the best (Table 2). However, the performance remained slightly inferior to that of the GLM for the cold samples (Figures 2 and 3), which is to be expected as the GLM was designed to better reconstruct cold climates.

For the MAT method, we used the chord distance on the 26 taxa. A maximum of ten analogues was selected. A threshold for exclusion of the analogues was calculated based on a Monte-Carlo method. Distance vectors between each modern spectra and another modern spectrum randomly selected were calculated 1,000 times for the entire matrix, and the threshold was the first quartile for these 1,000 values. In the case of the calibration data set, the threshold was calculated to 1.9. The reference data set used as a reserve of analogues was the calibration data set of 407 assemblages. The reconstructed values for the spectra in the four subsets of data were obtained from the weighted average of the SSTs of selected analogues according to the inverse of their distance. This method provided the best RMSE (Table 2), due to frequent geographical proximity of numerous samples. The RMSEP on the 406 samples remained narrow for the same reasons. More interestingly, we also obtained the best results for the cold and warm samples at the extremities of the modern data set (Figure 2), which cannot be explained by geographical proximity. This indicates that the method remains robust in a situation where the analogues are limited.

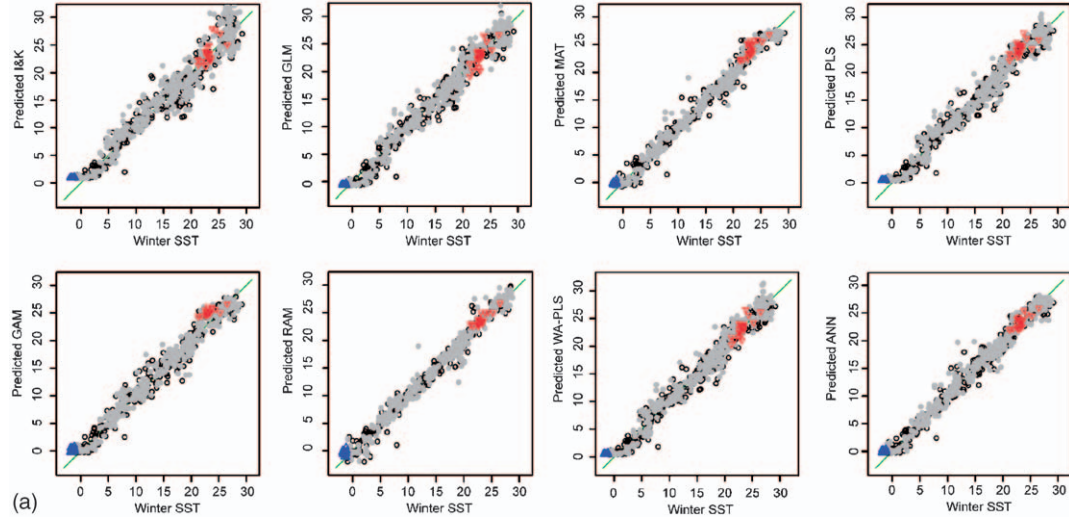


Figure 2 Cross-validation schemes showing the observed winter (a) and summer (b) SSTs (from [NODC, 1998](#)) against SSTs estimated by various methods (see calculated statistics in [Table 2](#)). The red and blue triangles correspond to the warm and cold samples, respectively; the open circles and grey circles correspond to the calibration and verification data sets, respectively.

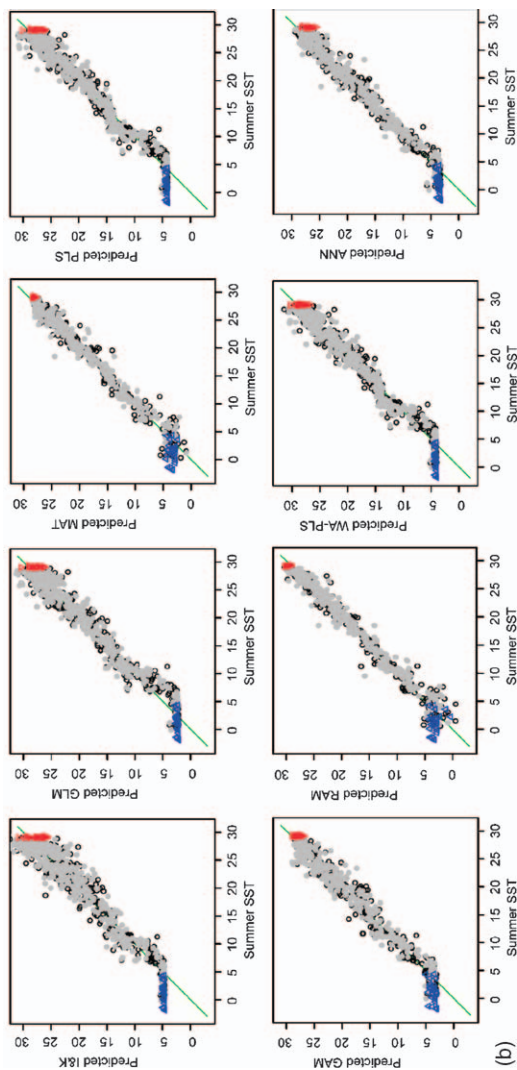


Figure 2 Continued

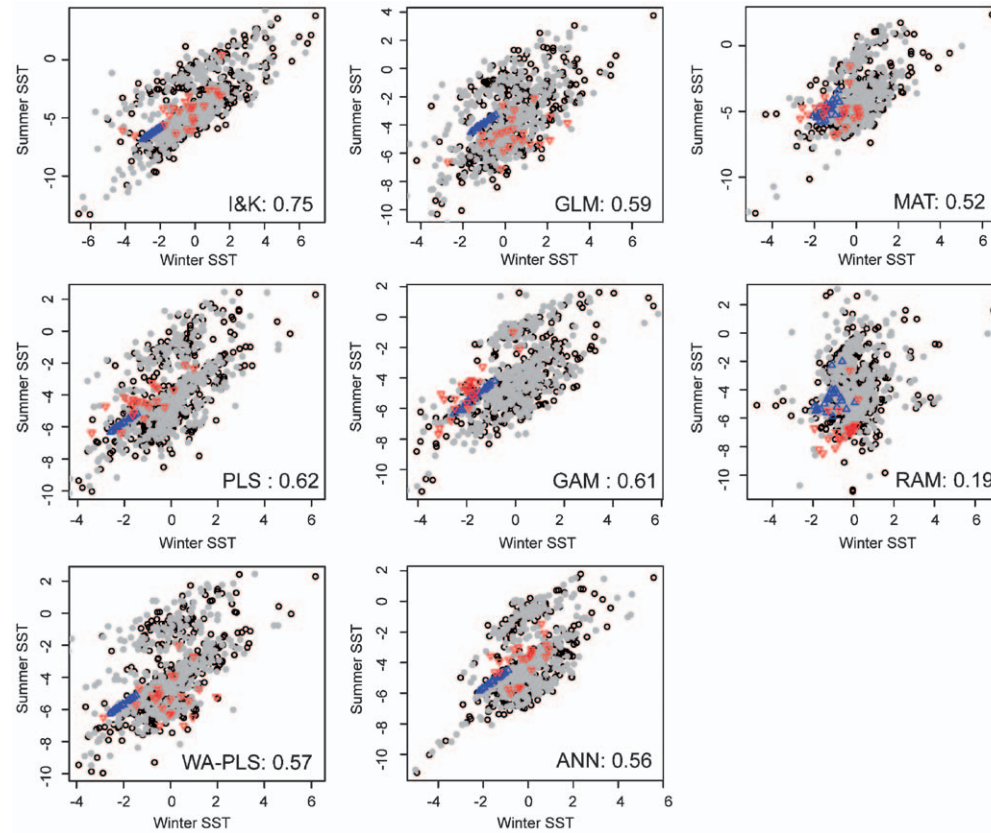


Figure 3 Winter vs. summer SST residuals (differences between observed and estimated values). The red and blue triangles correspond to the warm and cold samples, respectively; the open circles and grey circles correspond to the calibration and verification data sets, respectively.

For the RAM method, the modern data set was artificially expanded by remapping the original data onto a regular grid of SSTs (the climate of the new samples are represented by circles in [Figure 1B](#)). The 26 taxa were linked to the SST variables by a simple interpolation technique where the estimates of the taxa percentages at a grid point were weighted by the inverse of the distance between the grid point and the modern spectra in the climatic space. A maximum distance of 3°C was allowed. The ranges of SST variables were divided into 35 nodes spaced by 1°C, and the percentages of the 26 taxa were estimated on this grid, yielding 405 virtual samples (after removing samples falling outside the filled climatic space). MAT was then applied on the new set of spectra (real and virtual, i.e., 812 samples). We did not implement the “jump” technique ([Waelbroeck et al., 1998](#)), which enables a better control on the selection of analogues. Instead, we used a threshold, as in the MAT calculated with the Monte-Carlo method (in our case the threshold was 1.2). A better performance was obtained with RAM than with MAT, except for summer SSTs of the warm samples, which yield comparable results ([Table 1](#); [Figure 2a, b](#)).

5. DISCUSSION AND FUTURE DEVELOPMENTS

5.1. What are the Best Methods?

We have defined a general verification data set (406 samples) covering the same climatic space as the calibration data set. This was done to verify the risk of over-parameterization, which occurs when too many parameters are used, giving the impression that a large proportion of predictand variance is explained, while there is no real improvement of the prediction. In this respect, all of the methods provided reliable results, since the RMSEP on the general verification data set were close to the RMSE on the calibration data set. Because a clear spatial correlation exists between the assemblages, we cannot ascertain whether the RMSEP perfectly reflects the prediction capacity of each model. This is why we used cold and warm data sets. The cold data set represents the situation frequently met in the North Atlantic during glacial periods, with assemblages characterized by a low species diversity and the almost exclusive occurrence (>95%) of *Neoglobobulimina pachyderma* left-coiling (cf [Kucera et al., 2005b](#)). Conversely, the warm data set contains assemblages characterized by a high diversity of species. Both data sets are at the extremes of the reference data set: the cold data set represents winter SSTs lower than -1°C, whereas the calibration data set contains samples with winter SSTs higher than -1°C. The warm data set contains samples with summer SSTs higher than 29°C, whereas the calibration data set contains samples with summer SSTs lower than 29°C. This definition allows us to limit the spatial correlation and to verify the behavior of the models at the limit of the extrapolation, i.e., when there are no good analogues.

Examining [Table 2](#), which shows the respective performances of the methods, allows us to distinguish four categories:

- (1) The first category contains the method which performed the least well (I&K) on all the data sets, including the modern and fossil data (as shown by [Chen](#)

- et al., 2005). This was previously recognized by Pflaumann et al. (1996), Waelbroeck et al. (1998) and Malmgren et al. (2001).
- (2) The second category contains PLS, WA-PLS and GAM. They have a good performance, except for the cold and warm data sets. WA-PLS is nevertheless better than the two other methods for the warm data set.
 - (3) The third category of methods is represented by GLM, which makes it possible to weight the observations and to emphasize the importance of some samples for the optimization of the method in a given environmental range. This is what we have shown by focusing on the cold samples, with the result of more accurate reconstructions at the cold end of the data set. However, the corollary is a lesser performance for the opposite extreme, i.e., at the warm end of the data set.
 - (4) The last category includes the analogue methods (MAT, RAM) and ANN, which provide acceptable results for both extremes of the data sets, even if the performance is lower than that of GLM for the cold samples. From the analyses of lake diatoms, Racca et al. (2001) have already shown the superiority of ANN to WA and WA-PLS, and its robustness for the selection of species. In this category of methods, RAM outperforms MAT on the extreme data sets. The performance obtained by Waelbroeck et al. (1998) is apparently even better than ours for the cold end of the data set. This can be explained by the fact that, here, the coldest samples have been excluded from the reference data set, thus limiting the possibility of creating adequate virtual spectra in the domain colder than -1°C , whereas Waelbroeck et al. (1998) used all of the samples of the reference data set. For the warm samples, which are characterized by higher species diversity, this limitation had a lower impact. It can be argued that RAM creates virtual samples for combinations of ecological parameters that do not exist today, making the bold assumption that the response of the species within this space is smooth and continuous. If done with caution, this method offers somewhat of an advantage, because the gaps in the climatic space are often due to a lack of data rather than to ecological impossibilities. Nevertheless, these gaps must not be too large and numerous to enable a reliable interpolation.

The comparison of MAT and RAM has been made in a few paleoceanographical studies based on planktonic foraminifers. For example, Barrows and Juggins (2005) applied both approaches for the reconstruction of LGM SSTs along the Australian margin and the Indian Ocean; Chen et al. (2005) also tested and used these methods to estimate LGM SSTs in the western Pacific. Both studies concluded that MAT and RAM yielded results of similar quality. Hayes et al. (2005) used RAM and ANN for the reconstruction of the LGM SSTs in the Mediterranean Sea, also based on planktonic foraminifera. They concluded that the two techniques perform similarly in the LGM but that RAM would benefit from larger calibration data sets. Peyron and de Vernal (2001) have applied MAT and ANN on dinocyst assemblages from the Arctic Seas and the northern North Atlantic. They did not find any significant differences in the performance of both methods from the calibration and

validation exercises. However, they noticed that the application to late Quaternary series might yield different results in the case of distant analogue situations.

As pointed out by Kucera et al. (2005b), it is relevant to develop a multi-method approach, as it is not clear that any one method is indisputably better than the others. Kucera et al. (2005b) proposed a search for consistent patterns from the reconstructions based on different methods. A multi-method strategy was also suggested by Racca et al. (2001) who concluded that a better performance was obtained when estimates from WA-PLS and ANN are averaged, probably because the different methods capture and extract complementary information from assemblages. The comparison based on the example presented here shows that it is not necessary to try all of the methods, as some clearly underperform and others yield quite similar results. The best candidates for a multi-method approach are MAT, RAM, ANN and WA-PLS. GLM could be seen as an alternative to RAM.

5.2. What is the Effect of the Correlations between Climatic Variables?

The strong correlation between winter and summer SSTs ($r = 0.97$) in the reference data set makes it difficult to reconstruct seasonality patterns different from those occurring at present in the open North Atlantic Ocean, i.e., with a difference between summer and winter that is far from the mean present value of $3.9 \pm 0.15^\circ\text{C}$. In particular, the methods based on calibration tend to reproduce the structure that dominates the calibration data set. In Figure 3, we have plotted the winter vs. summer SST residuals obtained with the eight methods. The calibration methods (I&K, GLM, CAM, PLS, WA-PLS, ANN) result in significant correlation between the residuals of winter and summer SSTs. This indicates that a non-negligible part of the correlation between the seasonal temperatures is not related to the species assemblages, with the consequence that the species distribution is more diversified than the temperature pattern. There is then a risk that the equations will not be able to reconstruct a different temperature pattern. The MAT also results in some correlation, but not the RAM. By creating virtual samples, RAM has extended the climatic space and defined a limit for the distance between grid points and raw data of 3°C (Figure 1B), which gives more flexibility in the relationship between summer and winter. Because the virtual assemblages were created by using response functions of the species, there is probably no artifact. By re-sampling the response domain of the species, RAM creates coherent assemblages and provides better results.

5.3. How Many Taxa can be Used?

Racca et al. (2004) have discussed the selection of relevant taxa for diatom-based transfer functions in paleolimnology, which is a particularly delicate matter due to the extremely large diversity of diatom species and the large number of potentially determining environmental parameters in lakes. In order to limit the number of taxa (>250 species retained for statistical analyses; 109 surface sediment samples), Racca et al. (2004) used a pruning algorithm of ANNs. This is equivalent to a stepwise selection of predictors in a regression model. The removal of species that

corrupt the relationship between assemblages and environmental parameters make the reconstruction of the various parameters more independent. An approach like this may help to avoid a correlation structure such as the one found in Figure 3. However, the variables thus obtained independently could provide artificial combinations that do not exist in the present environment, which makes it difficult to provide a proper evaluation.

Generally speaking, some methods are more sensitive to large numbers of predictors. This is related to the parsimony principle, which stipulates that among several equivalent models, it is always better to choose the one having the lowest number of parameters. ANN is certainly the worst method in this respect. The first rule to follow is to remove the species that are proven not to be related to the variables to be reconstructed. The second rule is to preferably use orthogonal factors (principal components or correspondence factors) to restrict the number of parameters (this is preferable to the stepwise selection of predictors).

5.4. How Independent is a Cross-Validation?

As explained above, RMSE is not a proper measure of the quality of a method. It is simply a measure of the ability of the method to fit the training data set. Independent data must be used to independently calculate the RMSEP. However, the verification data set must be carefully defined. If the verification data set contains samples that are located geographically too close to the samples of the training data set, full independency of the data might not be achieved, which results in over-estimation of the quality of the reconstruction. For example, ANN is often optimized on a test data set to avoid over-parametrization; consequently, this data set loses its independence. Telford, Andersson, Birks, and Juggins (2004) have shown that SIMMAX and RAM fail to ensure statistical independency of samples during cross validation. They have also shown that the RMSEP of the best ANN or MAT models, as defined on the basis of cross validation, is lower than the RMSEP calculated from a fully independent test set. Moreover, ANN does not outperform MAT when independent test sets are used. Based on the analyses of the common example presented above, we offer the same conclusions.

Telford and Birks (2005), using planktonic foraminifer data, examined the consequence of spatial autocorrelation on the performance evaluation of WA, WA-PLS, MAT, and ANN. Spatial autocorrelation, which is the tendency of sites that are close to each other to bear more similarities together than with randomly selected sites, is an intrinsic property of ecological data; in the case of performance testing, spatial autocorrelation presents a problem because autocorrelated data violate the assumption of independence that is a prerequisite for most standard statistical procedures (Legendre, 1993). Telford and Birks (2005) showed that the coefficient of determination (r^2) between observed and estimated values from a transfer function model based on an autocorrelated environment can be high even in the absence of relationships between the species and the environmental parameter reconstructed. They showed that MAT and ANN outperform, in calibration, transfer function methods based on unimodal species-environment response models (calibrated on North Atlantic data), but that all methods yield similar results when validated on

spatially independent data (South Atlantic). They concluded that MAT and ANN could be misleading because of their incapacity to maintain a spatial autocorrelation structure. In our common example, truly independent data sets (the “cold” and the “warm” data sets) were used to take into account potential effects of spatial autocorrelation problems, but our results show that the conclusions based on these extreme data sets are not fundamentally different from the conclusions based on the validation data set obtained from random selection (the 406 samples data set in Table 2). Therefore, the conclusions of Telford and Birks (2005) are not unequivocal.

5.5. Bayesian Methods

The transfer function model uses a relationship which is not causal, as it expresses temperature as function of species abundances and vice versa. Recent development of Monte-Carlo-Markov-Chains (MCMC) methods has made accessible the use of causal models and their inversion in the context of Bayesian theory, which allows calculation of *a posteriori* probabilities for model parameters given *a priori* information and data. An example of application is given by Guiot et al. (2000) who used a mechanistic model of vegetation expressed as a function of climate. By inverting the model under the constraint that model outputs fit pollen data, they have reconstructed climate under various environmental conditions, such as low atmospheric CO₂ concentrations. The advantage of this approach is that observations under all conditions are not necessary, thus making it possible to reconstruct situations without analogue under the modern climate. Moreover, as the same vegetation model also simulates other useful parameters as $\delta^{13}\text{C}$ in soils, the same approach can be used to constrain the type of vegetation from $\delta^{13}\text{C}$ in loess profiles (Hatté & Guiot, 2005). This way of proceeding is possible on continents because global process models are available. It is not yet the case in the oceans but offers very promising possibilities.

In another example of Bayesian approach, Toivonen, Mannila, Korhola, and Olander (2001) have developed a multi-nomial calibration model, Bummer, which was applied to lake diatoms. In this case, there is no process model but rather a statistical model that is calibrated in a different way. They compared the Bayesian approach to a more standard approach called “frequentist approach”. The major difference lies in the fact that frequentist methods assume that parameters being estimated have fixed values and that data are random observations from some population, whereas the Bayesian approach assumes that parameters are random variables and that the data are fixed. In the same way, Haslett et al. (2006) propose the use of Bayesian theory and its MCMC implementation with a probabilistic model, where the discrepancies between data and estimates are modeled in a more sophisticated way. *A priori* probabilities allow some degree of smoothing to be imposed on the modern data in the spatial domain (it is achieved by the response surfaces) and on the fossil data in the time domain (it is achieved by imposing a relationship between time-neighbor fossil samples). The *a posteriori* probabilities provide the distribution of the climate, taking into account the fossil pollen data as well, in addition to the smoothness constraints. This methodology offers the possibility of working on the basis of several proxies: one proxy could be used for

defining *a priori* probabilities and another one to estimate the *a posteriori* probabilities. For example, in the context of paleoceanography, we may use the temperature reconstructed from planktonic foraminifera (with their confidence intervals) to define *a priori* probabilities, and the model calibrated on dinocysts or diatoms will be used to refine the reconstructions based on foraminifera. The final reconstruction is then compatible with both proxies. Such an approach has a great potential in paleoceanography for the development of integrated multi-proxy transfer functions.

6. THE APPLICATIONS OF TRANSFER FUNCTIONS *Sensu Lato* IN PALEOCEANOGRAPHY

6.1. The Most Common Methods and Reconstructed Parameters

In paleoceanography, the methods that have been the most commonly used are the transfer functions adapted from the Imbrie and Kipp (1971) method. They have been applied to planktonic foraminifer assemblages (e.g., CLIMAP, 1976, 1981; Ruddiman & Mix, 1993), radiolarians (e.g., Morley, 1979; Pisias et al., 1997; Abelmann, Brathauer, Gersonde, Sieger, & Zielinski, 1999), coccoliths (e.g., Molfino, Kipp, & Morley, 1982), and diatom assemblages from the Southern Ocean (e.g., Pichon et al., 1992; Zielinski, Gersonde, Sieger, & Fütterer, 1998; Gersonde et al., 2005) and the North Atlantic and Nordic seas (e.g., Koç & Schrader, 1990; Koç et al., 1993; Andersen et al., 2004). One advantage of I&K method was the possibility of developing transfer functions despite a limited number of modern spectra. This approach allowed important progress in paleoceanography during the 1970s and 1980s, when sedimentary cores were not very abundant. With the multiplication of coring expeditions, it has been possible to develop exhaustive databases of microfossil assemblages in surface sediment samples, and thus to adapt methods requiring large networks of reference samples, such as MAT and ANN. Since the early 1990s, MAT and methods derived from MAT (RAM and SIMMAX) are probably the most commonly used approaches; they were applied to foraminifer assemblages (e.g., Pflaumann et al., 2003; Chen et al., 2005; Kucera et al., 2005b), diatom assemblages (Crosta, Pichon, & Burckle, 1998; Crosta, Sturm, Armand, & Pichon, 2004; Gersonde et al., 2005), and dinocyst data (de Vernal et al., 1994, 2001, 2005; Marret, de Vernal, Benderra, & Harland, 2001). ANN is also becoming a method that is currently applied, but often as a complement of other methods (e.g., Kucera et al., 2005b; Chen et al., 2005).

The most common application of transfer functions using microfossil assemblages has been the reconstruction of the ocean surface temperatures of the past. Beyond general paleoclimate and paleoceanographic objectives, one of the objectives for SST reconstructions was to provide modelers with maps of the ocean temperature, which are as precise as possible, during past key intervals used in paleoclimate model experiments, notably within the context of COHMAP and PMIP (see COHMAP Members, 1988; Joussaume & Taylor, 1995, 1999). Since the pioneer work of CLIMAP (1976, 1981), the establishment of exhaustive

“modern” databases from analyses of surface sediment samples, together with the development of various methods, based on calibrations or similarities, has allowed important improvements in the ability to reconstruct past temperatures and to assess the degree of reliability of reconstructions (e.g., from the RMSEP; Section 4). A comprehensive series of publications on the use of various methods, based on several proxies (including foraminifers, siliceous microfossils and dinocysts), has been produced within the context of the MARGO project (see Quaternary Science Reviews, Vol. 24, issues 7–9 published in 2005; <http://www.pangaea.de/Projects/MARGO/>). In this special MARGO issue, regional maps of the LGM Ocean have been published and constitute updates from the LGM maps produced by CLIMAP in 1976 and 1981.

As mentioned above, the majority of paleoceanographic applications of transfer functions *sensu lato* aim at reconstructing SSTs, which is unquestionably the most determinant parameter of the distribution of planktonic organisms in the ocean. In general, the thermal inertia of the ocean waters results in a relative uniformity of the temperature throughout the annual cycle, which is reflected by the strong correlation between winter SST and summer SST. However, the temperature in surface waters can be a complex parameter at regional scales, with different seasonal signals or variants in the structure of the upper water masses (thermocline depth), depending upon stratification and vertical mixing. Thus, the temperature of the upper water mass may show a heterogeneous distribution in time and space. Some studies have provided analyses of the distribution of planktonic foraminifer assemblages and their relationships with the vertical structure of the upper water masses, leading to propositions of transfer functions for the reconstruction of the depth of the thermocline or mixed layer (e.g., Andreassen & Ravelo, 1997; Chen & Prell, 1997; Watkins & Mix, 1998; Huls & Zahn, 2000). Some studies based on the distribution of the mesopelagic radiolarians in the North Atlantic have led to the reconstruction of the thermocline structure from ANN (*cf.* Cortese, Dolven, Bjørklund, & Malmgren, 2005). Other researchers have examined the relationship between dinocyst assemblages and the SSTs of the warmest and coldest months of the year, which allowed them to propose reconstructions of various seasonal signals (*cf.* de Vernal & Hillaire-Marcel, 2000; de Vernal et al., 2001). These studies also suggested that the seasonal signal is intimately related to the stratification that is a function of salinity, and therefore led to the reconstruction of salinity and surface water density (e.g., Hillaire-Marcel, de Vernal, Bilodeau, & Weaver, 2001).

In addition to SSTs and related parameters, such as the depth of the thermocline or winter vs. summer temperatures, a few studies have examined the relationship between the density or seasonal duration of sea-ice cover and microfossil assemblages. They led to the application of MAT for the reconstruction of sea-ice cover at high latitudes of the Northern Hemisphere from dinocyst assemblages (e.g., de Vernal & Hillaire-Marcel, 2000; de Vernal et al., 1994, 2001, 2005), and in the Southern Ocean from diatom assemblages (Crosta et al., 1998, 2004; Gersonde et al., 2005).

Assuming a relationship between the trophic requirement of the species and primary productivity, diverse methods have been developed and tentatively applied to reconstruct biological productivity from planktonic foraminifer assemblages (e.g.,

Watkins & Mix, 1998; Cayre, Beaufort, & Vincent, 1999; Ivanova et al., 2003), dinocyst data (Radi & de Vernal, 2004) and diatoms (e.g., Schrader & Sorknes, 1991).

Attempts were also made to reconstruct diverse properties of sea water. In particular, Anderson and Archer (2002) proposed the reconstruction of carbonate saturation using modified MAT and planktonic foraminifer assemblages, which include some species more sensitive than others to calcium carbonate dissolution. Methods based on similarity have been applied to benthic ostracod data to tentatively reconstruct the bottom water mass properties, such as the pale temperature of shelf areas (e.g., Wood, Whatley, Cronin, & Holtz, 1993) or the origin of deep waters in the Arctic Ocean (e.g., Cronin, Holtz, & Whatley, 1994; Jones, Whatley, Cronin, Harry, & Dowsett, 1999). Many studies also explored the potential of benthic foraminifers to quantitatively reconstruct parameters such as productivity and carbon fluxes, bottom water quality and oxygenation, water depth, and deep current velocity, temperature and salinity. Because of the large number of species, complex ecological requirements and taphonomical processes, in addition to the typical mosaic pattern of the benthic foraminifer distribution on the seafloor, the development of transfer functions has been challenging (see Jorissen et al., this volume). To date, mostly indicator-species approaches have been used, but one can mention attempts at reconstructing paleo-water depth from MAT (Hayward, 2004), primary production and bottom water oxygenation using multivariate and regression methods (e.g., Loubere, 1994, 1996; Morigi, Jorissen, Gervais, Guichard, & Borsetti, 2001; Wollenburg, Kuhnt, & Mackensen, 2001).

6.2. Multi-Method and Multi-Proxy Approaches

Various methods and several proxies are now available for the reconstruction of paleoceanographical parameters, notably the SST. These methods include many techniques based on calibration or similarity, and various types of microfossils, but also molecular biomarkers (see Rosell-Melé & Clymont, this volume) and geochemical content of biogenic remains (see Rosenthal, this volume). Inter-comparisons of SSTs estimated from various methods are currently being made in order to improve the confidence level of paleoceanographical reconstructions.

An example of a down-core application of a multi-method approach is illustrated in Figure 4. It shows the results of MAT and ANN applied to dinocyst assemblages in a sequence of ~25,000 yr collected in the northwest North Atlantic (Peyron & de Vernal, 2001). In this example, three distinct calibration data sets, that yielded comparable RMSE from ANN, were used to reconstruct the SST of August, which is the warmest month. The results show very large amplitudes of temperatures from glacial to interglacial, with a good record of climate oscillations related to Heinrich and Dansgaard-Oeschger events (*cf.* Hillaire-Marcel & Bilodeau, 2000). In general, there is a relatively good agreement between the estimates given the extreme variability of conditions at the core site (in the hydrography used for calibration the modern SST in August was $13.4 \pm 2.6^\circ\text{C}$). The difference between the MAT and ANN estimates averages $1.5\text{--}2.6^\circ\text{C}$, which thus appear acceptable on the whole. However, SST discrepancies are particularly large for the early Holocene, which corresponds to an interval with relatively distant modern

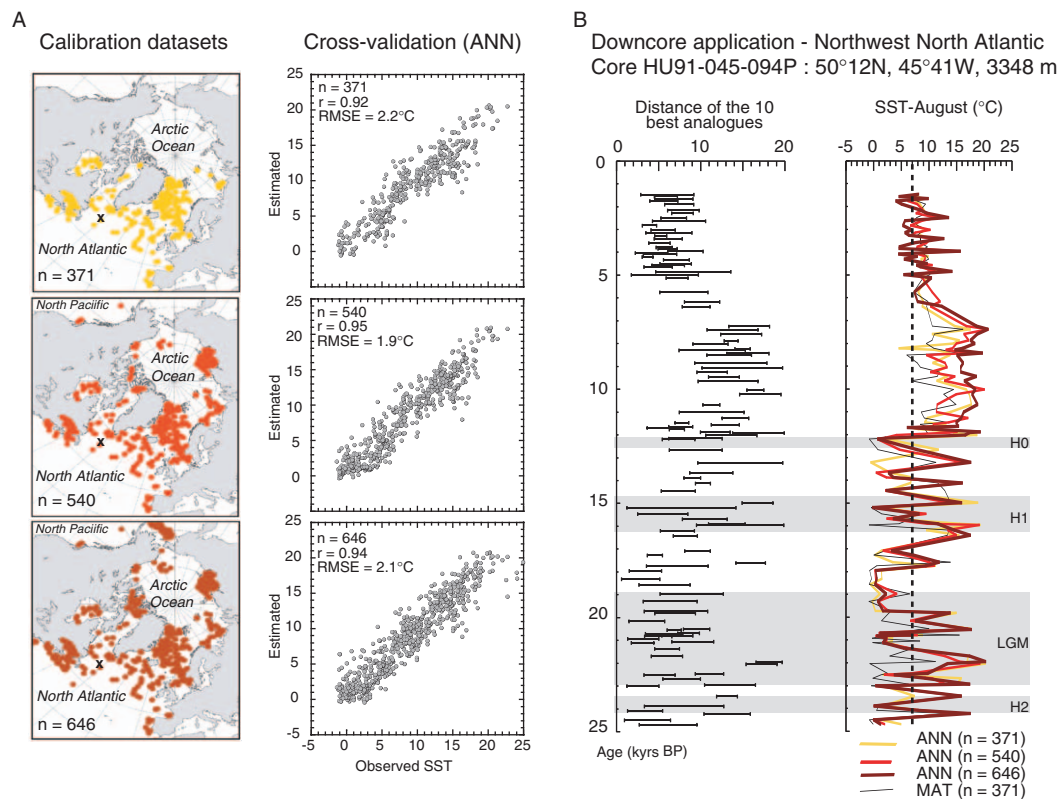


Figure 4 Data with permission from [Peyron and de Vernal \(2001\)](#) showing calibration results and down-core reconstructions of SST using ANN based on three dinocyst data sets, which represent the stages in the development of a single database that first covered the northern North Atlantic ($n = 371$) and progressively included the Arctic seas ($n = 540$) and the Northern North Pacific ($n = 646$). (A) Illustration of the calibration data sets in the left panel and of the cross-validation results for SST reconstruction in the right panel. The x on the maps show the location of the core analyzed. (B) The SST reconstructions from the three ANN calibrations are illustrated together with the results from MAT. The vertical dashed line corresponds to modern summer SST. It is noteworthy that most selected analogues are from the northern North Atlantic data set and that the MAT results are fairly similar among the three data sets since the selected analogues are approximately the same. The threshold for a no-analogue situation is ~ 40 .

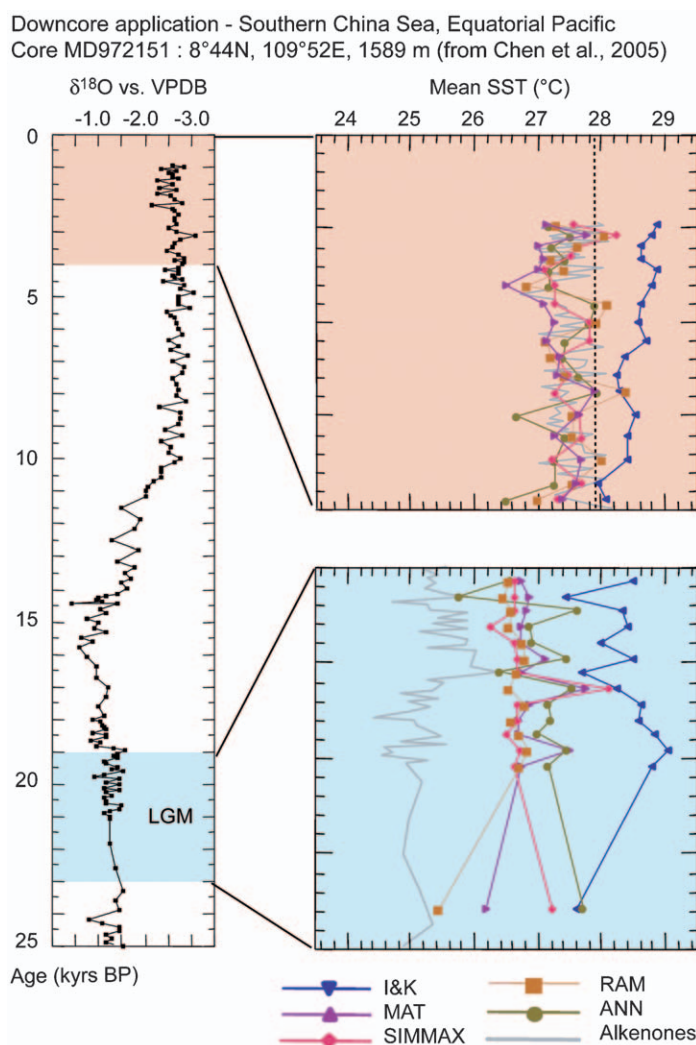


Figure 5 Data with permission from Chen et al. (2005, Copyright Elsevier) showing SST estimates in core MD972151 collected in the South China Sea (cf. Lee et al., 1999). The SST results are from five different transfer function methods using a western Pacific planktonic foraminifer data set that includes 694 samples. They are compared to SST estimates from alkenone data (Huang et al., 1999). The vertical dashed line in the Holocene part of the diagram corresponds to modern annual SST.

analogues (see Figure 5). From this example, it is clear that the definition of the calibration data set is determinant, since it allows for extrapolation and the divergence of the estimates. However, both calibration data sets yield satisfactory RMSE and the problem is to decide which one to retain.

Another example of a multi-method and multi-proxy approach, from a very different oceanographical setting, is shown in Figure 5. The data are from a

sequence spanning the late Holocene and the LGM in the South China Sea, equatorial Pacific (*cf.* Lee, Wei, & Chen, 1999; Chen et al., 2005). In this core, SST estimates were obtained from the application of various methods (I&K, RAM, SIMMAX, MAT, ANN) on planktonic foraminifers; independent SSTs from alkenone measurements (*cf.* Huang, Wang, & Zhao, 1999) were used for comparison. The western Pacific modern data set ($n = 694$) covers a wide range of temperatures (2–30°C) with a high density of points in the subtropical-equatorial domain. The cross-validation of all methods yielded RMSE ranging from 0.85 to 1.27°C, with the exception of I&K with a lower performance. Similarly, for the late Holocene interval, all methods provide consistent SST estimates in agreement with alkenone results, the only outsider being the I&K method. However, in this case, while SST estimates are consistent for recent time intervals, they yield discrepancies between alkenone and foraminifer proxies for the LGM and show a wide scatter of results in a few samples. This example illustrates that transfer functions, although well validated, cannot be widely extrapolated in space or time without some problems occurring.

The study of Chen et al. (2005) is one example, among others, that illustrates reconstructions of past intervals yielding significant discrepancies in SST estimates, i.e., with differences larger than the ranges of the RMSE. Many examples of differing reconstructions, or even conflicting SST estimates based on different proxies can be found in the literature (e.g., see Marret et al., 2001; Marchal et al., 2002). However, most often the comparison of the results is made on the basis of results obtained through individual projects, without standardization of the hydrographical database and transfer function methods. The formal inter-comparison exercise of the LGM that has been undertaken within the context of the MARGO project is probably a first in this direction. This inter-comparison has indeed implied first and foremost a standardization of the reference stratigraphy and of hydrographic data as input environmental variables (*cf.* Kucera, Rosell-Melé, Schneider, Waelbroeck, & Weinelt, 2005a). In the case of the MARGO inter-comparison exercise, the reconstructions have shown consistencies for many regions of the world oceans, which points to the adequacy of the methods used, and to the robustness of the quantitative results. However, in many regions of the ocean, in the northern North Atlantic, or along some continental margins marked by upwelling, the reconstructions show significant discrepancies (Kageyama et al., 2006; de Vernal et al., 2006).

Apparent inconsistencies recorded through multiple methods or multi-proxy approaches highlight the difficulty of making reconstructions in domains that represent extrapolation from the modern situation. Nevertheless, the inconsistencies themselves are important since they allow the identification of areas of uncertainties and lead to the identification of situations that are not well represented by the modern data sets used for paleoclimate reconstructions. SST is a relatively complex property of sea water that is intimately related to other parameters, including salinity, stratification and nutrients, pCO₂, etc. Although the temperature signal is most important, any change in the interaction among the properties of sea water may induce a bias in the SST record by a given proxy. Therefore, the discrepancies between SSTs estimated from various proxies could reflect the distinct response to changes in sea water properties that are interrelated in the modern data sets, but

possibly not in the same way as in the past (e.g., Watkins & Mix, 1998; Mix et al., 2000). As expressed by Mix et al. (2000), “a multi-proxy strategy increases confidence in results and provides oceanographic context for more effective interpretation of processes”.

7. CONCLUDING REMARKS

Transfer functions, in a broad sense, are powerful methods that allow complex biological assemblages analyzed by micropaleontologists to be translated into simple abiotic parameters for the use of the entire scientific community working in the fields of paleoceanography and paleoclimatology. They are extremely useful, but they imply a high degree of simplification of multiple biotic variables that are themselves dependent upon complex abiotic and biotic parameters. Various methods were developed over the last decades, based either on calibrations between assemblages and climate parameters or on the degree of similarity between assemblages (see Table 1). Among these methods, those yielding the most accurate reconstructions of the modern conditions are the weighting average — WA-PLS method, the ANN, the MATs and RAM. The artificial neuronal method is, however, very dependent upon the definition of the calibration data set, which makes it more risky than others for extrapolating outside its domain of training, because of its strong non-linear characteristic. The analogue methods do not necessarily provide accurate results outside the domain of calibration, but they allow reliability indices to be set based on the degree of analogy, as well as the identification of non-analogue situations. The RAM provides the possibility of improving the reference data sets in some parts of their range, but we must be cautious not to create virtual samples that are too distant from the actual climatic space. It is probably one of the most appropriate and reliable methods currently used in paleoceanography.

A multi-method approach has been proposed to obtain more robust reconstructions and to identify the assemblages for which the performance of the transfer function is low or problematic. Our comparison shows that it is not necessary to try all of the different methods, as some clearly underperform and others yield quite similar results. The best candidates for a multi-method approach are MAT, RAM, ANN and WA-PLS. The remaining problem is to make a decision from the various reconstructions. Should we select one, which is assumed to be the “best”, or average all estimates? The first solution is difficult to apply, as we generally do not know which one is the “best” reconstruction. Averaging several reconstructions, however, may be biased by the inclusion of outliers, thus leading to an average different from the initial reconstructions. We can advise researchers to make at least three reconstructions, eventually remove possible outliers, and, finally, to use average values or a median, which is less sensitive to outliers.

Another classical problem of transfer functions comes from the possible correlation between the environmental variables to be reconstructed. It is necessary to collect surface samples representative of diverse environmental conditions in order to limit this problem; otherwise it is difficult to make an inference about a given

parameter if conditions of the past were different from the present. In this respect, we have seen that the RAM method performs better than the other methods. Another strategy to evaluate is to select an appropriate subset of species for each environmental variable analyzed. Such a “stepwise” approach based on different predictors may help in producing calibration equations for each predictand that are more independent.

The respective performances of the transfer function methods can be evaluated from the cross-validation of modern data sets. For example, when tested on modern planktonic foraminifer data in the North Atlantic, the best methods produce temperature estimates within $\pm 1.3^{\circ}\text{C}$, and those performing less well produce estimates within an error of $\pm 2.2^{\circ}\text{C}$. These performances are possibly overestimated because of autocorrelation (*cf.* Telford & Birks, 2005), but contrary to what is claimed by these authors, MAT and RAM are not more affected by this problem than are other methods. In any case, a performance with a RMSEP of $\pm 1.5^{\circ}\text{C}$ should be considered extremely satisfactory since the actual standard deviation of the “modern” mean SST is $>1.5^{\circ}\text{C}$ in the largest part of the World Ocean, and throughout most of the North Atlantic (see *World Ocean Atlas*, 2001).

Beyond the evaluation of the performance of various methods, one important caveat is the fact that the parts of modern data sets that serve as analogues or as a basis for calibration do not necessarily provide the adequate environmental combinations of ocean surface properties for the past we seek to reconstruct. This issue has to be explored through multi-proxy strategies and would benefit from the development of Bayesian approaches, based on either process or stochastic models.

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